

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 207 — Past Questions & Materials
by Renance

Course: MTH 207
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

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Paper 1 of 1 — MTH 207 Real Analysis 1 2018 2019 First Semester

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EBBURUN!

DEPARTMENT OF SCIENCE & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

SEMESTER EXAMINATION 2018/2019

COURSE: MTH 217 — Real Analysis I

INSTRUCTIONS: Attempt ALL questions in PART A and any 2

questions in PART B. Time Allowed: 2'4 Hours

PART A = (MULTIPLE CHOICE QUESTIONS)

1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2 + 3x + 3$. Which of the following is true?

(a) f is continuous at $x = 3$ given that $\epsilon > 0$ (b) f is not continuous at $x = 3$ given that $\epsilon > 0$

(c) f is continuous at $x = 3$ (d) f is not continuous at $x = 3$

2.

Find the range of $f(x) = \sqrt{1-x}$ defined on $[-1, 1]$.

(a) $y \in [0, 1]$ (b) $y \in (0, 1)$ (c) $y \in [0, 1)$ (d) $y \in (0, 1]$

3. Find the limit $\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$ where $f(x) = \sqrt{1-x}$. (a) $x \in [-1, 1]$ (b) $x \in (1, 2)$ (c) $x \in (-1, 1)$ (d) $x \in (1, 2]$

4. Find the limit $\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$ where $f(x) = \sqrt{1-x}$. (a) $x \in [-1, 1]$ (b) $x \in (1, 2)$ (c) $x \in (-1, 1)$ (d) $x \in (1, 2]$

5.

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \sqrt{1-x}$. Which of the following is true?

(a) f is continuous at $x = 1$ (b) f is not continuous at $x = 1$

(c) f is continuous at $x = 1$ (d) f is not continuous at $x = 1$

6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \sqrt{1-x}$. Which of the following is true?

(a) f is continuous at $x = 1$ (b) f is not continuous at $x = 1$ (c) f is continuous at $x = 1$ (d) f is not continuous at $x = 1$

7. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \sqrt{1-x}$. Which of the following is true?

(a) f is continuous at $x = 1$ (b) f is not continuous at $x = 1$ (c) f is continuous at $x = 1$ (d) f is not continuous at $x = 1$

8. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \sqrt{1-x}$. Which of the following is true?

(a) f is continuous at $x = 1$ (b) f is not continuous at $x = 1$ (c) f is continuous at $x = 1$ (d) f is not continuous at $x = 1$

9. Which of the following sets is an interval on the real line $\mathbb{R}$? (a) $\{x \in \mathbb{R} : x \leq 3\}$ (b) $\{x \in \mathbb{R} : x < 3\}$ (c) $\{x \in \mathbb{Z} : x < -20\}$ (d) $\mathbb{N}$

10. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \sqrt{1-x}$. Which of the following is true?

(a) f is continuous at $x = 1$ (b) f is not continuous at $x = 1$ (c) f is continuous at $x = 1$ (d) f is not continuous at $x = 1$

11. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \sqrt{1-x}$. Which of the following is true?

(a) f is continuous at $x = 1$ (b) f is not continuous at $x = 1$ (c) f is continuous at $x = 1$ (d) f is not continuous at $x = 1$

12. Which of the following sets contains its infimum? (a) $\{x \in \mathbb{R} : x \leq 3\}$ (b) $\{x \in \mathbb{R} : x < 3\}$ (c) $\{x \in \mathbb{Z} : x < -20\}$ (d) $\mathbb{N}$

13. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \sqrt{1-x}$. Which of the following is true?

(a) f is continuous at $x = 1$ (b) f is not continuous at $x = 1$ (c) f is continuous at $x = 1$ (d) f is not continuous at $x = 1$

14. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \sqrt{1-x}$. Which of the following is true?

(a) f is continuous at $x = 1$ (b) f is not continuous at $x = 1$ (c) f is continuous at $x = 1$ (d) f is not continuous at $x = 1$

both infimum and supremum (d) None of the above

$(-\infty, -10)$ (b) $\{x \in \mathbb{N} : x \geq 1\}$ (c)

(a) 4 (b) 1 (c) 4.414 (d) none of the above

contains its supremum only (C) contains

17) Which of the following sequences is convergent? (a) $\frac{1}{n} - \frac{1}{n^2}$ (b) $\{((-1)^n)^{1/n} : n \in \mathbb{N}\}$ (c) $\int_0^1 x^n dx$ (d) None of the above

(8) Which of the following sequences is $n(-1)^n : n \in \mathbb{N}$ (b) $((2^n - 3)7^n : n \in \mathbb{N})$ (C)

$\{x : x \in \mathbb{Z}\}$ (d) None of the above

7 bounded? (a) $\{$

$\frac{1}{n}, \text{ there exists } d > 0 \text{ s.t. } f(x) \text{ at } x = 2$ (a) $< -$ (b) < 25 (c) < 5 (d) $< =$

$||, : 7] :$

20. Suppose that $0 < \epsilon$, compute $\delta > 0$ if $f(x) = 2x + 7$ is continuous at $x = -2$ (a) > 1 (b) < 1 (c) $> (d)$

> 2

for $y \in (0, 1)$ (by $y \in [0, 1]$) $\forall \epsilon \in [9, 1]$

71. Find the range of $f(x) = \sqrt{1 - x^2}$ defined on $\mathbb{R}$.

(a) $\mathbb{R} - \{0\}$ (b) $\mathbb{R} \cup \{0\}$ (c) $[-1, 1] \cup (0, 2]$ (A) $(2, 0)$ (0,]

function may

92. Find the domain of $f(x) =$

23. If a function is continuous on a bounded interval (a, b) , the range of that interval under the function

is (a) unbounded (b) bounded (c) bounded from below (d) all of the above

74. If a function is continuous on a bounded interval $(a, b]$, the range of that interval under the function

may be (a) unbounded (b) bounded from below (d) all of the above

19. For all $x \in \mathbb{R}$, $f(x) = -5x + 3$ is continuous

on a closed interval under the

function (c) bounded

25. Find the first derivative of $f(x) = \sin x$, $x \neq \frac{\pi}{2}$ (a) $-\cos x$ (b) $-\sin x$ (c) 0 (d) $\cos x$

26. $\lim_{n \rightarrow \infty} (\frac{1}{n^2} + \frac{1}{n})$ (a) 0 (b) $\frac{1}{2}$ (c) $\sqrt{3}$ (d) ∞

77. Find the limit of the sequence $\frac{1}{n^2}$ (a) 0 (b) $\frac{2}{3}$ (c) ∞ (d) $-\frac{1}{2}$

28. The set $\{n \in \mathbb{N} : n \leq 10\}$ is (a) Not bounded (b)

bounded

Find the limit of the sequence

(c) only bounded below (d) bounded

$\lim_{n \rightarrow \infty} \frac{1}{n^2} :$

converging? (a) $\frac{1}{n}$ (b) $\{n\}$ (c) ∞ (d) None

(4, 5) (b) ∞ (c) $\lim_{n \rightarrow \infty} \frac{1}{n^2}$ (2, 10).

only bounded above $\{$

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99. Which one of the

lx for which $2x - 6$

30. Find the values O